

Lecture 3 - Aug 31

Permutations Practice Problems

Practice 1

How many ways are there to arrange 26 lowercase letters so that no two vowels are consecutive?

Answer

$$S = \{a, b, \dots\}$$

permutations of S such that vowels don't share spaces

First arrange 21-non vowels: $21!$ ways.

Basically finding # ways of inserting vowels between non-vowels (guarantees no consecutive). In this case 21 positions are filled, meaning 22 open spaces to insert into:

$$21! * P(22, 5) = 21! * \frac{22!}{(22 - 5)!}$$

Practice 2

How many ways are there to seat A, B, C, D, E, F around a circular table, if A refuses to sit next to B ?

Method 1: Subtraction principle

ways to place 6 people: $\frac{6!}{6} = 5!$

ways to place A and B together:

- First merge A and B into one person: $\frac{5!}{5} = 4!$
- Then, account for both combinations A, B and B, A : $4! * 2$

$$\frac{6!}{6} - 2 * \frac{5!}{5} = 5! - 2 * 4! = 3 * 4!$$

Method 2: Direct counting #1

First sit A as head of the table (1 way). Then, place on the left and right of A (4 choices and 3 choices, respectively). Then we are left with 3 seats that can be filled in any ways, linear permutation:

$$1 * 3 * 4 * 3! \\ 3 * 4!$$

Method 3: Direct counting #2

First sit A as head of the table (1 way). Then, place down B , 3 choices (not next to A or on top of A). Then, place

the rest (linear permutation):

$$\frac{1 * 3 * 4!}{3 * 4!}$$

Method 4: Direct counting #3

ways to seat only $B, C, D, E, F \rightarrow$ circular permutation:

$$\frac{5!}{5} = 4!$$

Now place A in one of the gaps, but cannot place in the gaps next to B , so 3 gaps to place in:

$$3 * 4!$$

Combinations

Let $r \geq 0$ be an $\mathbb{Z}$ and let S be a set with $|S| = n \geq r$.

An r -combination of S is an unordered selection of r objects of the n elements in S .

An unordered collection of elements from a set is also the definition of a **subset** $\rightarrow$ Combinations and subsets are equivalent.

Remark: An r -combination is an r -element subset of S

$$\binom{n}{r} := \# \text{ r-combinations of } S$$

Combination Identities / Special cases

$$\binom{n}{0} = 1, \text{ only one way to pick zero elements: } \emptyset$$

$$\binom{n}{n} = 1, \text{ only one way to pick the whole set}$$

$$\binom{n}{n-1} = \binom{n}{1} = n, \text{ taking/choosing one element from } S$$

$$r > n \rightarrow \binom{n}{r} = 0, \text{ no way of choosing more elements than exist}$$

Combinatorial proof: a proof that uses counting arguments to prove a theorem or a statement.

Theorem 1

For $0 \leq r \leq n$,

$$\binom{n}{r} = \frac{P(n, r)}{r!} = \frac{n!}{r!(n-r)!}$$

Proof: We only have to prove that

$$P(n, r) = r! * \binom{n}{r}$$

Let S be an n -element set. Each r -permutation of S arises in exactly one way as a result of carrying out the following two steps:

1. choose/select r elements of from S
2. arrange the chosen r elements

ways to carry out 1. is $\binom{n}{r}$ (by definition)

ways to carry out 2. is $r!$ (number of ways to permute $r!$ elements).

By multiplication principle,

$$P(n, r) = \binom{n}{r} * r!$$

Theorem 2.3.2:

For $0 \leq k \leq n$.

$$\binom{n}{k} = \binom{n}{n-k}$$

Proof Method 1: Algebraic proof (not allowed on tests!)

$$\begin{aligned} \frac{n!}{n!(n-k)!} &= \frac{n!}{(n-k)!(n-(n-k))!} \\ &= \frac{n!}{n!(n-k)!} \end{aligned}$$

Proof Method 2: Combinatorial proof

Let $S = \{a_1, \dots, a_n\}$, be an n -element set

For any $A \subseteq S$ w/ $|A| = k$

$$X = \{A \subseteq S \mid |A| = k\}$$

$$\begin{aligned} |\bar{A}| &= |S| - |A| = n - k \\ Y &= \{B \subseteq S \mid |B| = n - k\} \end{aligned}$$

$$\text{Then } |X| = \binom{n}{k}, |Y| = \binom{n}{n-k}$$

Define $f : x \rightarrow y$ as follows:

$$\forall A \in X, f(A) = \bar{A} \in Y.$$

Then f is a bijection from x to y only one compliment of a set.

It follows that

$$\binom{n}{k} = |X| = |Y| = \binom{n}{n-k}$$

□

Theorem 2.3.3

For $1 \leq k \leq n-1$:

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

*Remark: This is called **Pascal's identity**.*

Intuition

For every element a we have two choices: whether to choose it or not

1. if we *don't* choose a , then we still have to fill k choices, however now we have one less element to choose from (a).
2. if we *do* choose a , then we only have to fill $k-1$ choices, and since we took a we also have one less element to choose from.

Proof

Let S be an n -element set and pick $a \in A$. Count # k -element subsets of S :

Define $X := \{A \in S \mid |A| = k\}$

Then $|X| = \binom{n}{k}$

Note for understanding

Here X_1 is all subsets (size k) that include the element a , and X_2 is all the subsets that don't include element a . Together they form a partition of X .

Now partition X into X_1 & X_2 :

$$X_1 = \{A \subseteq S \mid |A| = k \cap a \in A\}$$

$$X_2 = \{A \subseteq S \mid |A| = k \cap a \notin A\}$$

Note for understanding

In X_2 we have to fill k spots, however we know $a \notin A$ meaning we cannot consider a as an option in our subset.

Then $|X_2| = \# k\text{-element subsets of } S - \{a\} = \binom{n-1}{k}$.

Note for understanding

In X_1 we have to fill k spots as well, however we are guaranteed one of them will be filled with a , so we only really need to fill k . We also don't have access to a as a choice since it was already chosen, so n gets reduced by one here as well

Then $|X_1| = \# (k-1)\text{-element subsets of } S - \{a\} = \binom{n-1}{k-1}$.

By the addition principle:

$$|X| = |X_2| + |X_1| = \binom{n-1}{k} + \binom{n-1}{k-1}.$$